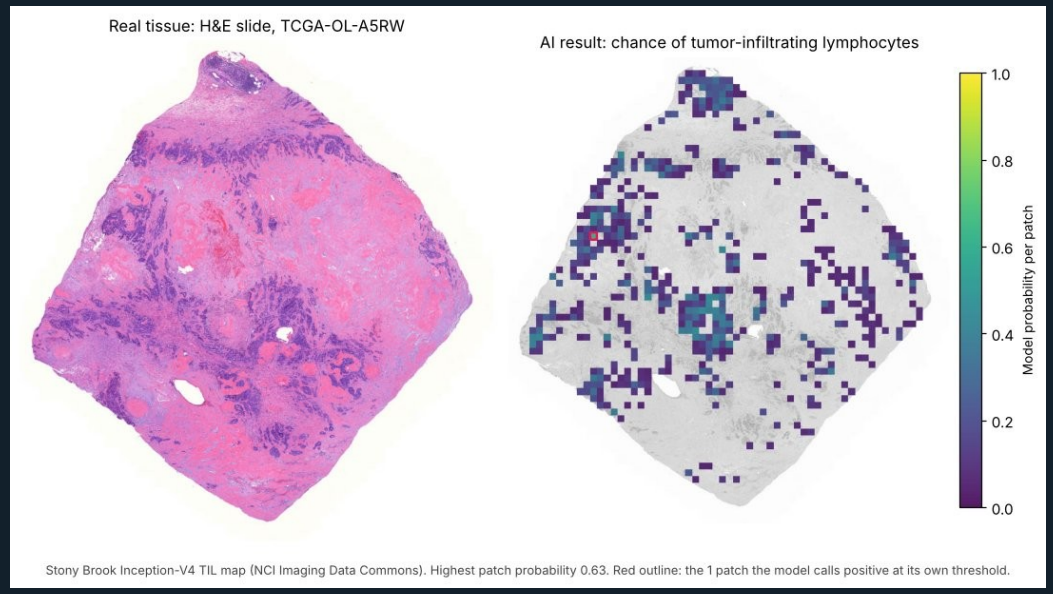
HEALTHCARE AI HACKATHON · KICKOFF

OncoSys Cancer AI Lab

Build AI tools that help cancer researchers trust their data and decide what to do with it.



Wednesday, November 11 · 7:00 PM ET
The Open Accelerator · Red Hat · Northeastern University · OncoSys AI

TONIGHT

40 minutes, then your questions

Time (ET)	What
7:00	Welcome and why connecting cancer evidence is hard
7:10	The challenge: tracks, six featured products, the data
7:20	How judging works, what to submit, the rules
7:27	Live: run the starter in Colab
7:32	The week ahead, teams and links
7:40	Questions

WHY THIS MATTERS

Real cancer evidence doesn’t always line up

535

of 1,101 breast tumors in TCGA have every data layer. Each join changes who is in the study.

11%

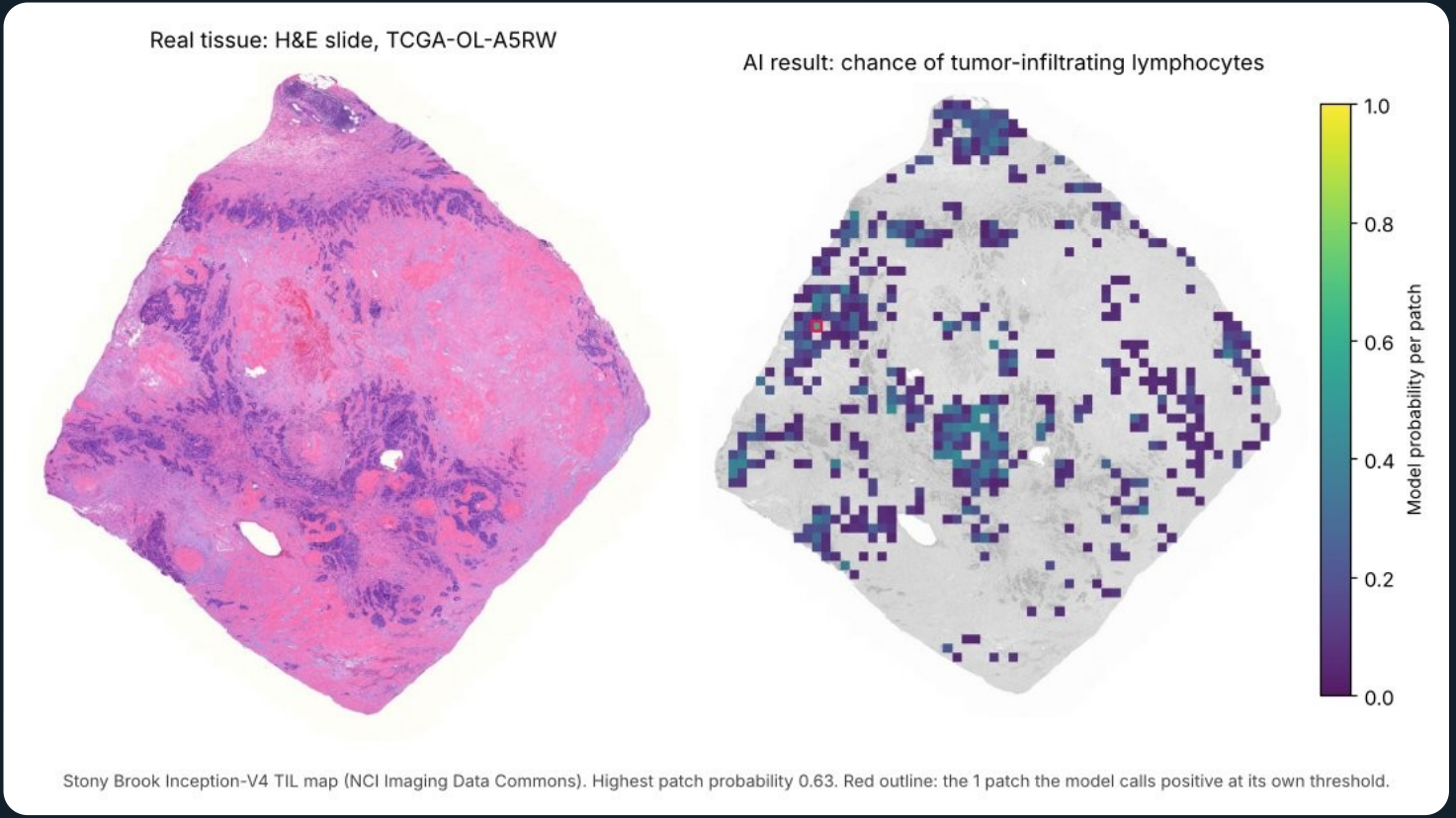
of tumors get a different molecular subtype depending on which call you use.

1,978

of 3,111 tissue slides are frozen sections, not diagnostic slides.

Source: TCGA-BRCA via UCSC Xena and NCI Imaging Data Commons; counted for this event, November 2026.

Is this AI result right, and how would anyone check?



Stony Brook Inception-V4 TIL map (NCI Imaging Data Commons). Highest patch probability 0.63. Red outline: the 1 patch the model calls positive at its own threshold.

Left: a real tumor slide from TCGA.

Right: a published model's estimate of where immune cells are.

Its highest patch score is 0.63, and at its own threshold it calls **one patch** positive.

THE CHALLENGE

Pick a user.
Build one workflow they can finish.

Give a judge **one result they can verify** and **one limitation they can inspect**.

TWO TRACKS

Catch the problem, then help someone decide

Cohort Firewall

Catches what could make a result unreliable.

Must prove: which samples can answer the question, and how a change in the data affects the result.

Evidence Navigator

Helps someone investigate a result and decide what to do.

Must prove: a reviewer can trace a claim to the data, reproduce it, and see where it stops being supported.

SIX FEATURED PRODUCTS

Pick one, combine two, or pitch your own

Cohort Control Tower

Compare two data releases; rerun only what changed.

Cohort Studio

Build a cohort from plain English; catch joins that double-count.

Multimodal Case Room

One case: slides, record and molecular data, sources linked.

Evidence Copilot

Answers questions by running real analyses, with sources.

Research Claim Auditor

Checks each claim in a report against the data.

Repair Review Desk

Reviews proposed data fixes; approve, reject or ask for more.

Teal: Cohort Firewall · Pink: Evidence Navigator · Briefs, practice inputs and ideas: products/ in the repo

THE DATA

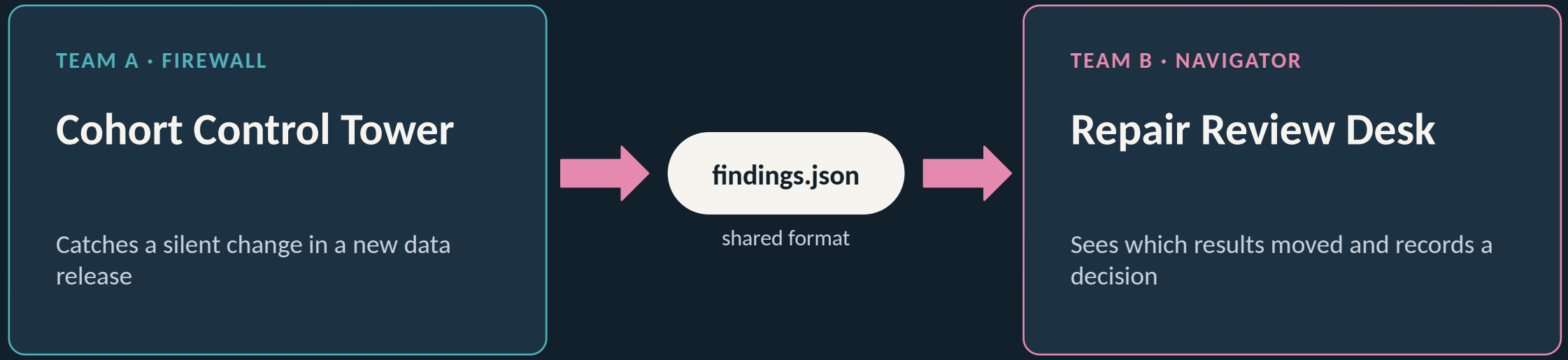
Start small; go bigger only if you need to

Pack	One record is…	Laptop start
TCGA breast cancer (core)	A tumor sample: clinical, survival, DNA, RNA, protein, slide links	~70 MB
PathMNIST	A labelled colorectal tissue patch	Smallest version
CPTAC	A lung, colon or kidney case, with protein data and slides	Under 1 MB
LIDC-IDRI	A chest CT with radiologists' nodule marks	34–230 MB
DepMap	A lab-grown cancer cell line	~100 kB
Trial records	A trial's eligibility rules, with fictional cases	Small
CAMELYON17	A lymph-node slide from one of 5 hospitals	Labels only

Only join records with a documented match. Tissue patches, cell lines, tumors and CT scans describe different things.

THE FINALE

Two teams’ tools, working as one



Live, on data neither team has seen. We’ll pair Control Tower and Repair Review Desk teams tonight.

HOW JUDGING WORKS

Judges use your tool on something you haven’t seen

1

3-minute demo

Your user, your workflow, one result and one limitation.

2

A hidden test

In the 2 minutes of questions, you get a new input and run your tool on it live.

3

A quick check

For four products, a script compares your output file with the answer, and judges see a short summary.

4

7-criterion rubric

Each track has its own 100-point rubric, scored by five judges.

A working core beats an ambitious half-build. And “this data can’t answer the question, and here’s why” can win.

CODE FREEZE · SATURDAY 2:15 PM ET

What to submit

Everyone

- A repo judges can open, with a README: product, user, workflow, setup steps
- Runs from a fresh clone (ask a teammate to try it)
- Uses the frozen TCGA release; extra data listed
- A short demo video link
- No API keys or secrets in the repo

Output files the checker reads

Product	File
Cohort Control Tower	findings.json
Repair Review Desk	decisions added
Research Claim Auditor	audit.json
Cohort Studio	cohort_samples.csv

Check JSON files with format/validate.py.

THE RULES

Real patient data, research use only

No clinical claims

No diagnosis, treatment advice or risk scores. A tissue label on an image is a research output.

No re-identification

The data is de-identified but came from real people. Don’t link it to outside records about individuals.

AI that shows its work

Run real queries, cite sources, say when the data can’t answer. Use a local model or the user’s own key.

Models are welcome

Test on held-out patients, slides or cell lines, and add a short model card: data, split, where it fails.

LIVE NOW

Run the starter in 15 minutes

1. Open the repo and click Run the starter (Colab)
2. It downloads the TCGA data and checks every file
3. See what one row means
4. Watch the cohort shrink as you add data layers
5. Open a real tissue slide in your browser

REPO

github.com/disleveK/oncosys-hackathon-challenge

No GitHub account needed to open it. To run it in Colab, sign in to Google.

No Google account? Clone it, **pip install -r requirements.txt**, and open starter/tcga_starter.ipynb in Jupyter.

THE WEEK AHEAD

Four days, two checkpoints, one finale

Wed · tonight

Kickoff. Form your team, pick a product, run the starter.

Thu · build

Checkpoint 1: post one working result in Discord.

Fri · build

Checkpoint 2: swap repos with your paired team; each reruns the other from the README.

Sat · finals

In person at the Cabral Center, Northeastern. Every team member attends.

Saturday (ET)	What happens
9:30 AM	Doors open, breakfast
10:00 AM – 2:15 PM	Overview, then build time with lunch; **code freeze at 2:15 PM**
2:30 – 4:00 PM	Demos: 3 minutes plus 2 minutes of questions
4:30 – 5:00 PM	Finale and awards

TEAMS AND HELP

Register your team, then find us in Discord

Register your team

Captains only: **red.ht/toa-oncosys-team-matching**

Teams of 3–5. Register by **Thursday, 1:30 PM ET** with your team name, members and track.

Get help

Discord: **red.ht/oncosys-discord**

Find teammates, ask mentors, share checkpoints. Email: hackathon@the-open-accelerator.com

Who’s helping

10 • 4

10 industry judges and 4 mentors from Red Hat on Saturday.

Questions?

1 · Explore the data

2 · Pick a mission

3 · Run the starter

All three start at github.com/disleveK/oncosys-hackathon-challenge
Help: red.ht/oncosys-discord · See you Saturday at the Cabral Center.